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Construction of approximate reasoning model for dynamic CPS network and system parameter identification

Juxia Xiong *, Jinzhao Wu

Chengdu Institute of Computer Application, Chinese Academy of Sciences, Chengdu, Sichuan, 610041, China
University of Chinese Academy of Sciences, Beijing 100049, China
Guangxi Key Laboratory of Hybrid Computational and IC Design Analysis, Guangxi University for Nationalities, Nanning, Guangxi, 53006, China

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ABSTRACT

CPS (Cyber Physical System) is a large and complex real-time feedback system that integrates computing processes, physical processes, communication networks, sensor networks, and control systems. It has the powerful function of sensing and controlling the physical environment, which is a big wave following the Internet technology. Because the forms of communication, interaction, and collaboration between heterogeneous units within the CPS are intricate and complex, a comprehensive model needs to be established to describe and analyze the CPS. This paper analyzes the CPS architecture and proposes a new and more complete CPS architecture, decomposes according to this architecture, and classifies the physical entities in the CPS. At the same time, event-based modeling thinking is used to define, classify and formalize event messages. Considering the higher real-time requirements of CPS, an event weighting algorithm was designed according to the different priorities of real-time events. In order to reduce the congestion caused by the limited network bandwidth in the CPS system, improve the ability to identify abnormal data with great uncertainty, and fully guarantee the response rate of the CPS system to emergencies, this paper analyzes the complexity of the CPS system from the perspective of information theory. The average dynamic complexity of the CPS system is set as a threshold to determine the level of information entropy of the sensor data in a certain period of time. The CPS system selects high information entropy data to send first. The effectiveness is analyzed through experiments.

1. Introduction

With the rapid development of computer technology and network communication technology, computing and communication capabilities will soon be embedded in all objects and structures of the physical environment, and people's attention will also be more focused on the relationship between various computing elements and physical elements [1,2]. CPS (Cyber Physical System) is a complex system that completes the integration of people, machines, and things based on the Internet of Things [3]. It realizes the interconnection between information time and physical time. The information world detects and manages the physical world through sensing technology and integrates it. The design between physical entities, computing entities and interactive entities is ensured, and the system is completed efficiently and in real time, providing people with more convenient, intelligent and secure services [4]. In the CPS system, wireless sensors are widely distributed in various monitoring areas, and they are numerous. They are many-to-one data transmission methods with the network, but the bandwidth of the network is very limited, which often causes congestion.

Since the CPS was proposed by the National Science Foundation in 2006, it has attracted a lot of attention from domestic and foreign experts and scholars in the past few years [5,6]. CPS has become one of the new commanding heights of international information technology competition, and is considered to be the third important milestone after the computer and the Internet [7]. Relevant scholars divide the research work of CPS into the development design, performance requirements, and application areas of CPS based on the existing results, and give an introduction separately, which lays the foundation for further research by development designers [8]. The deployment and application of CPS in security-related areas such as intelligent transportation, aerospace, and defense make it extremely demanding for safety and reliability. The inaccurate description of the real model by the system model will directly affect the operation of the later system [9]. Therefore, before the CPS system is put into use, it must undergo strict analysis, testing and sufficient verification. In terms of formal modeling and verification of CPS, researchers at home and abroad have achieved some results [10–12]. These include CPS modeling methods based on Actors and Agents, modeling and verification methods based on hybrid automata, differential dynamic logic, and

* Corresponding author at: Chengdu Institute of Computer Application, Chinese Academy of Sciences, Chengdu, Sichuan, 610041, China.
E-mail address: xiongjuxia1107@163.com (J. Xiong).

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Petri nets [13]. Agent-based modeling uses some inherent characteristics of the agent, such as behavioral autonomy, environmental coordination, interaction of functions, and structural distribution, to characterize the fusion of computing and physical processes. Relevant scholars use Agent to model the intelligent water conservancy system in CPS applications, and achieve the purpose of real-time network control of water flow and its distribution [14,15]. Researchers propose a CPS modeling and attribute verification framework based on differential dynamic logic [16]. CPS was first modeled using Hybrid UML, and then the universal model was converted to a Hybrid Program based on the principle of semantic consistency. Combined with the Hybrid Program to verify the system attributes to be verified, finally the theorem prover Ke Ymaera was used to automatically verify the attributes. The disadvantage of this method is that the conversion between models needs to be further improved [17]. Ontology models are usually used to model event contexts [18,19]. However, the traditional ontology model cannot accurately model the uncertainty of the event, which makes it not widely used in the processing of uncertain events. However, the event context has a positive effect on the processing of uncertain events. Relevant scholars have proposed an ontology modeling and context-sensitive method, which collects contextual knowledge fragments from different ontology and integrates them into standardized, clear abstraction levels and modules [20–22]. The CPS model may use different development languages, and the data format to be processed is also different. Therefore, we must first solve the problem of interoperability of different data between these heterogeneous models. Moreover, the communication protocols used between models may be different, so achieving efficient communication between models is an issue that must be considered during the modeling process.

This paper analyzes the CPS architecture and proposes a more complete CPS architecture. It also classified and formalized the physical entities and event messages in the CPS. An event weighting algorithm is designed according to the different real-time priorities of events. For the completeness of the system data and the scientificity and accuracy of the data analysis and processing, this paper supplements the data with low information entropy. At the data receiving end, the information with low information entropy uploaded according to the time interval T. It is estimated through input interaction, filter calculation, probability model update, output interaction, etc., combining Kalman filtering and Bayesian prediction ideas. The validity of this method is verified by experiments.

The rest of this paper is organized as follows. Section 2 studies the formal expression of event-based CPS. Section 3 studies the network data system model under CPS. Section 4 summarizes the full text and points out future research direction.

2. Formal expression research based on event CPS

2.1. CPS architecture

Since the concept of CPS was put forward, various aspects of CPS research have also begun, including the initial design of the CPS position network structure. Because CPS emphasizes the interaction and integration between computing, communication, control systems, and the physical environment, and at the same time has adaptive capabilities, the design of the CPS posture network structure also varies.

In the study of CPS, the marriage structure of the position system is the cornerstone and the guarantee of its characteristics. Therefore, this paper analyzes the characteristics and principles of CPS and the synergy and integration of each module, and proposes a new and more complete CPS posture system marriage structure, as shown in Fig. 1.

This architecture has the following main parts:

(1) Sensor network: It consists of several sensor program points and collection nodes. Sensor nodes are used to obtain information about the physical environment and physical entities. At the same time, the sensor nodes will sense the situation after the execution of the command and send feedback to achieve a closed loop of the entire behavior event. If a certain attribute of the obtained physical object exceeds a certain degree, the sensor node will generate a corresponding CPS simple event and pass the collected physical attribute and the generated event to the collection node. On the one hand, the collection node fuses the same physical attributes collected from different sensor nodes or the same physical attributes collected at different times from the same sensor node. If the fusion attributes exceed a certain limit, a corresponding fusion event is generated, and the physical attributes, simple events and fusion events are released to the information center and stored as historical data, while the simple events and fusion events are released to the control center for further processing. On the other hand, the aggregation node will also forward sensor commands from the control center to the corresponding sensor nodes.

(2) Control center: The control server is responsible for processing and analyzing simple events and fusion events from the sensor network, or combining the events according to the predefined logic to generate corresponding composite events. Correct control can be generated through internal logic decisions. Thus, control commands are issued to the control nodes of the actuator network, and the composite events and control commands generated are issued to the information center.

(3) Information center: The server based on mathematical data is responsible for storing historical data such as physical attributes, simple events and fusion events from the sensor network, as well as composite events and control commands from the control center.

(4) Actuator network: It consists of a number of actuator nodes and control nodes. The control node is responsible for receiving control commands and distributing the commands to the corresponding actuators for execution in order to control certain physical attributes of the physical objects and achieve changes in the physical environment.

(5) User terminal: A specific terminal device is the interface between the operating user and the CPS. Users can receive information required for inquiries or issue artificial control commands on these terminal devices.

(6) CPSRTnet: The CPS real-time network is a network system that connects all modules in the entire architecture and is a guarantee for information transmission.

2.2. CPS main modeling methods

CPS uses the three basic operations of "Computation, Communication, and Control" to realize the interaction between the physical world and the information world through trusted, controllable, and extensible networked physical devices to achieve specific functions. The CPS system model is shown in Fig. 2.

(1) Agent-based integrated modeling

The Agent has autonomy and responsiveness, and can observe the environment by sensing itself and knowing itself in the environment that is defined by itself. The effect of the effector on the environment is very similar to CPS.

The integrated modeling of CPS based on Agent uses the autonomy, interaction, coordination and structural distribution of Agent to describe the fusion between computing and physical processes.

Agent-based modeling method can use the existing Agent programming language and modeling tools, which has certain convenience. However, the agent's autonomy can also make it make decisions to a certain extent while satisfying the conditions. This may bring unpredictable management problems to the CPS model, and it is also not conducive to the safety and reliability of the CPS. On the other hand, only the behavioral specifications of individual agents are given, but no specifications for the dynamic behavior of the system are given. Therefore, Agent-based CPS modeling methods also need to consider the unpredictable uncertainty and the linguistic representation of system dynamic behavior.

(2) Integrated modeling based on Hybrid System

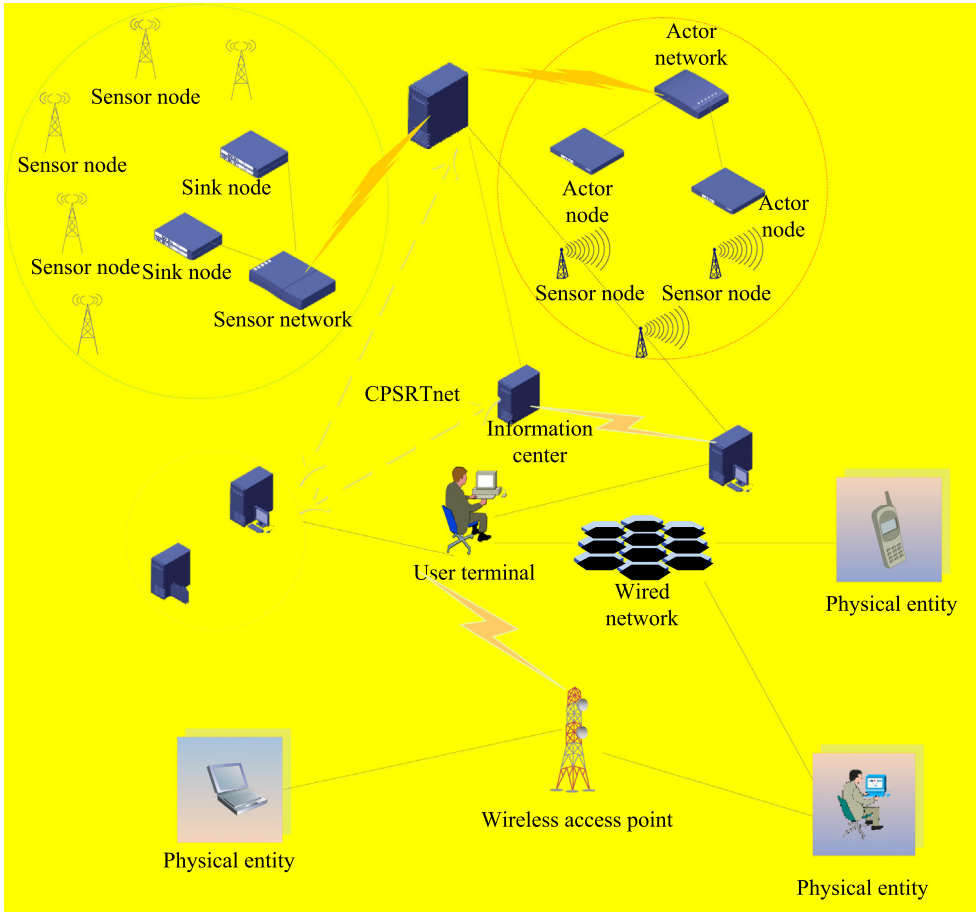


Fig. 1. Schematic diagram of CPS architecture.



Fig. 2. CPS system model.

Hybrid System is a subclass of real-time embedded systems, which contains the situation where discrete logic jumps are mixed with continuous real-time behavior. Because it is a type of system that has both discrete and continuous behavior characteristics, the hybrid system can also combine the physical process based on the connection duration and the calculation process based on the finite state machine.

The hybrid system-based modeling method can undoubtedly effectively solve the problem of the integration of continuous physical processes and discrete computing processes of CPS. However, most general hybrid system models have not considered the real-time interaction between physical processes and computing processes in the network. Therefore, there is a need for expansion and more detailed analysis.

(3) Event-driven modeling method

The event-driven modeling method defines the content of interaction between various behavioral entities in CPS as event behaviors. Events here not only refer to the behaviors of physical entities during their activities, but also include the physical entity's attribute information and related Co-signal. In the model, corresponding response events must be generated for events to obtain coordinated control of the physical environment. For large-scale and complex systems such as CPS, event-driven thinking is a very effective method.

The event-driven CPS modeling method well addresses the needs of CPS for spatiotemporal elements and integration of heterogeneous systems. This paper will improve and optimize based on the event model, and propose a new and complete CPS event model.

2.3. CPS physical entity modeling

In the CPS event model, each physical entity must have some basic characteristics:

(1) The physical entity in the CPS needs to have a unique identity ID in the entire system. This ID is used as the unique identifier of the physical entity to distinguish it from other physical entities.

(2) In addition to the identity ID, the physical entity also needs to have uniform space–time attributes and state attributes. The time attribute of the physical entity refers to the time point at which the physical entity is currently located in a unified time axis. The spatial attribute refers to the coordinate position of the physical entity in the unified spatial coordinate system or the location information of the area in the physical environment. State attributes refer to the state of the physical entity itself, such as stationary, moving, idle, busy, etc.

(3) The sensing device has the ability to sense environmental information, and can send the information to the control server through a network medium.

(4) The execution equipment can receive the control instructions transmitted by the control server through the information network, and execute the instructions correctly according to the instructions.

Here, by analyzing the characteristics of physical entities in CPS, physical entities with the same attributes and behaviors are abstracted and defined as physical entity classes.

We define an Entity Class as a triple:

$$EnC = (EC_{id}, CP, OP) \tag{1}$$

Among them, the EC_{id} in the triple represents the physical ID of the physical entity class, which is used to uniquely identify the physical entity class; CP represents a two-dimensional set of all attributes of the physical entity class; OP represents the set of behavior of this physical entity class. These behaviors are well-defined and vary according to different applications and different physical entity classes.

For each physical entity that exists in the CPS, it will belong to a physical entity class name, and each physical entity is an instance of a physical entity class. The physical entity is defined here as a five-tuple:

$$En = (HT, EP, EOP, EnC, E_{id}) \tag{2}$$

Among them, E_{id} represents the identity ID of the physical entity and is used to uniquely identify it in the CPS; EnC represents the physical entity class to which the physical entity belongs; EP refers to the attribute set of the physical entity, EOP refers to the behavior set of the physical entity, and HT refers to the hybrid space–time Petri net in which the physical entity is located.

2.4. Event message modeling

In CPS, in addition to the formal expression of the most basic physical entities, it is also necessary to formalize the event messages passed between the physical entities. The CPS model designed in this paper is based on the idea of event-driven, that is, the messages transmitted between various parts of the CPS and various physical entities are expressed in the form of events. Event messages can not only represent changes in some attributes, but also can convey some information.

Because CPS is an event-based system, all calculation, communication, and control operations are based on event units. The processing of events can be completed by processing their attribute characteristics.

Based on the inherent properties of the event, and considering the characteristics of the event composition and spatial distribution, this paper designs a new event representation method.

$$E = (e, P, T, O, w, L_0, T_0, L) \tag{3}$$

e = (eid, t), eid represents the ID of the event, which is generated at the same time when the event is generated, and is the unique identifier of the event in the entire system. t represents the type of event, and can also be reflected at the subscript of E.

P represents the attributes of the event, in the form of a collection, $P = \{p_1, p_2, \ldots, p_n\}$. T represents the time when the event was generated. L represents the place where the event was generated. T_0 represents the time when the observer found the event. L_0 represents the place where the observer found the event. O stands for the observer, that is, the sensor node that discovers the occurrence of the event. w represents the weight value of the event to reflect the real-time level of the event, which will affect the scheduling and processing of the event.

According to the classification of events, simple events are denoted as Es, fusion events are denoted as Ef, and compound events are denoted as Ec. For an event message, the attribute values included are not equally important, and different attributes have different requirements for the real-time event. Therefore, the attribute real-time requirements need to be reflected by the attribute weight. It is assumed here that the weight of each attribute is known. The contribution of each attribute to the event is expressed as g (p_i), and the weight value of the event is calculated by the following formula:

$$w_{E_j} = \sum_{i=1}^{n} g(p_i) \tag{4}$$

A brief description of the event-based weighting algorithm is shown in Fig. 3.

3. Research on network data system model under CPS

3.1. Interactive multi-model algorithm

The interactive multi-model algorithm can track the target well. Compared with the Kalman filtering algorithm, it is a sub-optimal estimation of the system state, which is suitable for running on a computer and has a small amount of calculation. The interactive multi-model algorithm selects the corresponding model to match the target according to the actual movement form of the target, which can be a single model or multiple models. The most common models are the CV model and the CS model. The CV model represents the constant speed model. The CS model represents the acceleration model. Each model has a corresponding filter, and each filter is independent of each other. Each filter uses Kalman filtering algorithm for filtering, and an

```
Receive event
                for
Each event type
      do
                for
    Each attribute of the event
      do
 Calculate the contribution of the attribute to the event g (pi) based on the known attribute weight
                end for
 Calculate event weight value according to formula
                end for
 Select the event corresponding to the maximum weight value for processing
controller   generates decision commands based on events
 Modify environment value
```

Fig. 3. Brief description of event-based weighting algorithm.

interactive effector corresponds to the input end of the filter, and the output end corresponds to an estimation mixer. With the change of the target motion form, the models will be transformed accordingly. The probability of transformation between the models can be expressed by Markov matrix. The interactive estimation of the filter is based on the model's probability and the transformation between the models through the interactive effector. According to the probability computer, the interactive multi-model algorithm calculates the state estimate of each model filter based on the interactive estimates of the filters and the observation data, and finally calculates the weighting from the filters of each model in parallel, and then obtains the output results.

The interactive multi-model algorithm includes a four-step cyclic process of interaction between models, filtering prediction, model updating, and output results. It is a recursive algorithm.

3.2. Model construction

We suppose there are k models in the CPS system environment:

$$X_t = GZ_t + Y_{t-1}X_t \tag{5}$$

Among them, X_t is understood as the state of the CPS system at time t, Y_{t-1} is understood as the state transition matrix of the CPS system, which can represent the process of CPS system state transition in two consecutive time periods, and G is understood as the interference transfer matrix. Covariance is a white noise sequence with matrix Q and mean 0. The mutual conversion of k models can be realized with the help of Markov Chain. Its transition probability matrix is:

$$P = \begin{bmatrix} p_{11} & M & p_{1k} \\ N & O & N \\ p_{k1} & M & p_{kk} \end{bmatrix} \tag{6}$$

Measurement model of wireless sensor is:

$$W_t = U_t + H_t X_t \tag{7}$$

W_t is understood as the measurement value of the wireless sensor, and H_t is understood as the observation matrix of the CPS system. It can represent the relationship between the data value obtained from the wireless sensor at the data acquisition end of the CPS system and the state transition of the CPS system.

As can be seen from the content of this paper, there are static and dynamic complexity in the CPS system environment. The state and number of states that occur depends on the data obtained from the sensors and the scheduling scheme of the corresponding system. The state is static complexity, and the actual state that is not as expected is dynamic complexity. Deviation from this premise is expected to switch from static complexity to dynamic complexity. Therefore, the predictable model and unpredictable model of the CPS system can be derived from the two target tracking model classes of the CV constant speed model and the CS acceleration model in the interactive multi-model algorithm.

If it is assumed that the data in the CPS system is two-dimensional data, X (n) represents the state of the CPS system, and X (n) is determined by the node location, static complexity, and dynamic complexity. If the sensor data collection cycle is T, the conversion speed from predictable complexity to unpredictable complexity is λ, then the dynamic model and observation model in the Cartesian coordinate system are:

$$X(t+1) = FX(t) + GV(t) \tag{8}$$

$$Z(t) = W(t) + H(t)X(t) \tag{9}$$

Among them, t represents the time point when the sensor collects data.

(1) Predictable model

The state transition matrix is:

$$Y = \begin{bmatrix} 1 & T & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & T \\ 0 & 1 & 0 & T \end{bmatrix} \tag{10}$$

The interference transfer matrix is:

$$G = \begin{bmatrix} \frac{T^2}{2} & 1 \\ 0 & 0 \\ 0 & \frac{T^2}{2} \\ 1 & T \end{bmatrix} \tag{11}$$

The observation matrix is:

$$H = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{12}$$

(2) Unpredictable model

The state transition matrix is:

$$Y = \begin{bmatrix} 1 & \frac{\sin(\lambda T)}{\lambda} & 0 & 1 \\ 0 & \cos(\lambda T) & 1 & \sin(\lambda T) \\ 0 & 1-\cos(\lambda T) & 1 & \frac{\sin(\lambda T)}{\lambda} \\ 0 & \sin(\lambda T) & 0 & \cos(\lambda T) \end{bmatrix} \tag{13}$$

The interference transfer matrix is:

$$G = \begin{bmatrix} \frac{T^2}{2} & 1 \\ 1 & 0 \\ 0 & T \\ 0 & \frac{T^2}{2} \end{bmatrix} \tag{14}$$

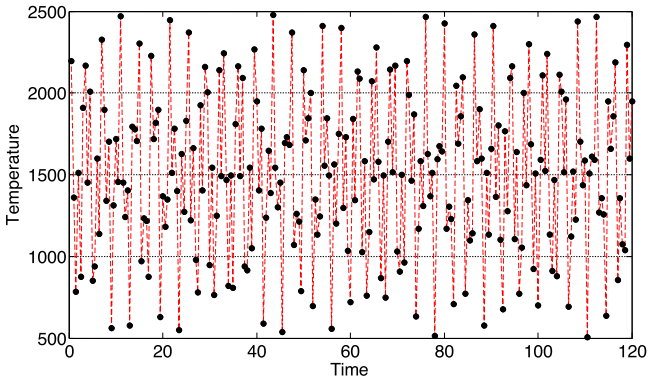


Fig. 4. True information entropy low data distribution.

The observation matrix is:

$$H = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \tag{15}$$

3.3. Experiments and results analysis

In order to verify the method of complementing low information entropy data described in this section, this paper takes the monitoring of temperature sensor data by a furnace monitoring system based on CPS in the industry as an example. The furnace monitoring system transmits the information collected by the temperature sensor every 30 s. All real information collected within 120 min of low entropy data is distributed, as shown in Fig. 4.

Now on the basis of all the low information entropy data in 120 min, we select low information entropy data at intervals of 30 s, and then use these low information entropy data to obtain predicted values by the method in this paper. Assume that the original node position coordinates (x, y) of the data in the CPS system are (1200, 1200), the static complexity is (20, 20), and the period for the sensor to collect data is T = 30 s. We observe x and y separately.

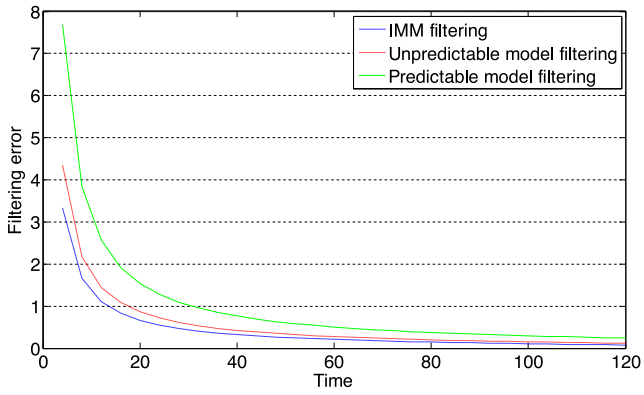
When using the interactive multi-model algorithm for filtering calculations, two sets of models, predictable model and unpredictable model, are used. Now we assume the state shift rate of the unpredictable model, and the Markov Chain transition matrix is:

$$P_1 = \begin{bmatrix} 0.97 & 0.01 \\ 0.01 & 0.97 \end{bmatrix} \tag{16}$$

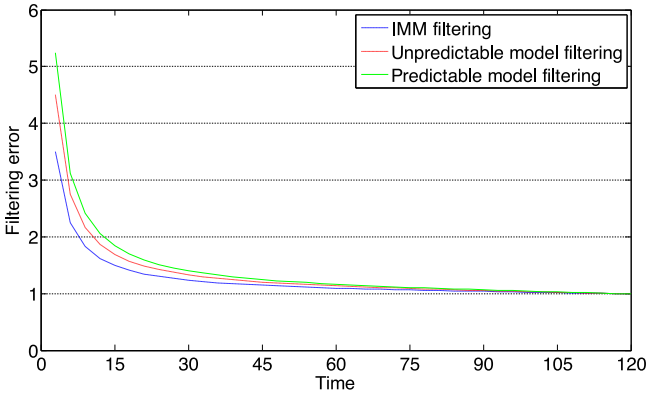
In order to make a directional analysis of the filtered prediction results, we use the predictable model, unpredictable model, interactive multi-model filtering of the X and Y node positions to find the node position deviation and root mean square error, respectively, as shown in Fig. 5 and Fig. 6. The predictable and unpredictable model probabilities for each time point are obtained accordingly, as shown in Fig. 7. It can be found from the figure that at the time of state migration, the probability of the unpredictable model is greater than the predictable model probability. At this time, the key of interactive multi-model filtering is affected by the unpredictable model. This is the same as usually expected. The interactive multi-model algorithm is to complete the prediction of low information entropy data in the CPS system through the automatic adjustment of the probability of each model. Compared with a single model filtering, its prediction accuracy is better. In the figure, IMM stands for interactive multi-model.

The low-entropy data prediction results obtained by the method in this paper within 120 min are shown in Fig. 8.

According to the above simulation experiment on the low information entropy data selected within 120 min, it can be concluded that the key of the interactive multi-model algorithm is to filter and predict the low information entropy data. This is a comprehensive calculation process, during which it can automatically switch and adjust models,



(a) X-direction filtering error



(b) Y-direction filtering error

Fig. 5. Node position filtering deviation.

based on the low entropy of information selected at 120 s intervals within 120 min. The data can well predict all the low information entropy data within 120 min.

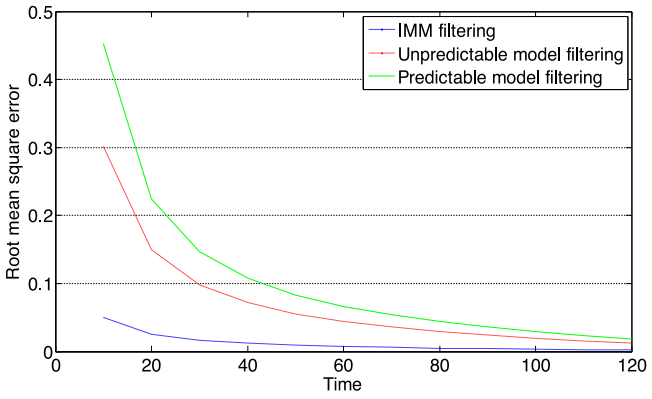
Next, let us take a look at all the low information entropy data within 120 min, select low information entropy data every 30 s, and then use Kalman filter algorithm to predict these low information entropy data. The prediction result is shown in Fig. 9.

In order to further compare the predicted value of Kalman filter with the predicted value of this method, this paper introduces the root mean square error as the evaluation index.

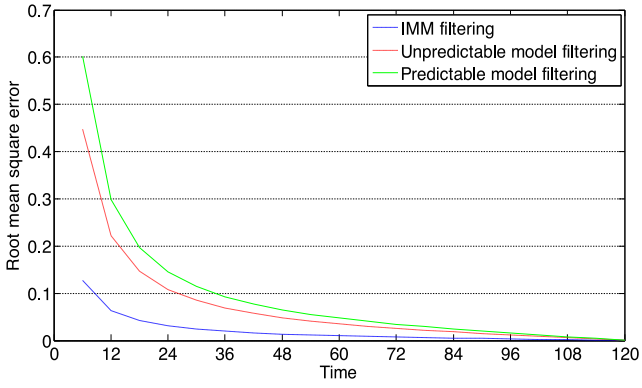
In Fig. 10, by comparing the predicted value of low information entropy data with the root mean square error of the actual value, it can be concluded that the prediction of the method with low information entropy data is more accurate than that of Kalman filtering.

4. Conclusion

As a new research area, CPS has many problems that need to be solved. Because the forms of communication, interaction, and collaboration between heterogeneous units within the CPS are intricate, a comprehensive model needs to be established to describe and analyze the CPS. This depends on good modeling methods and comprehensive simulation and verification tools. Because CPS is a deeply embedded real-time feedback system, this paper proposes a more comprehensive approach after considering the two aspects of the system's interaction with the external physical world, the processing of internal information, and the transmission of control commands. In order to reduce the congestion caused by the limited network bandwidth in the CPS system, improve the ability to identify abnormal data with great uncertainty,



(a) Root mean square error in X direction



(b) Root mean square error in Y direction

Fig. 6. Mean square error of position filtering.

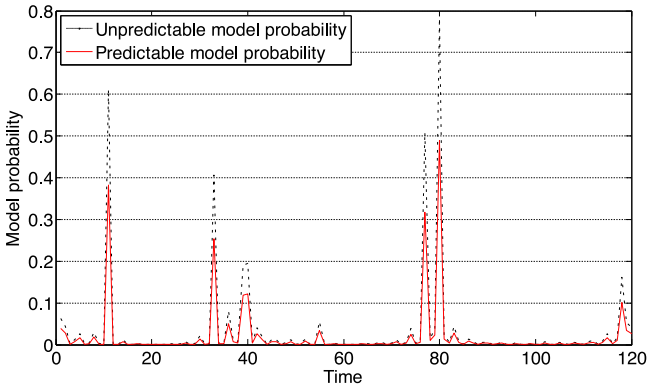


Fig. 7. Predictable and unpredictable model probabilities at each time point.

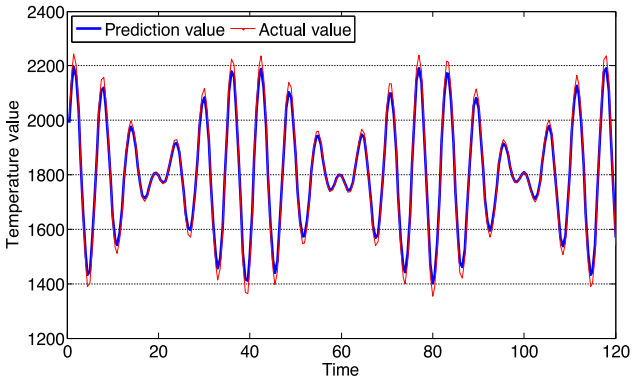


Fig. 8. Results of the predictions of our method.

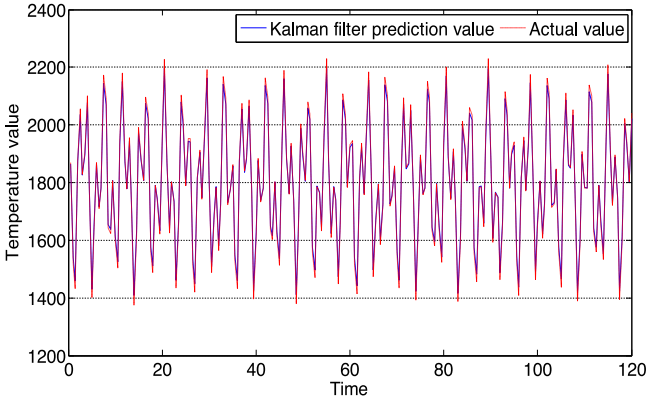


Fig. 9. Results of Kalman filtering prediction.

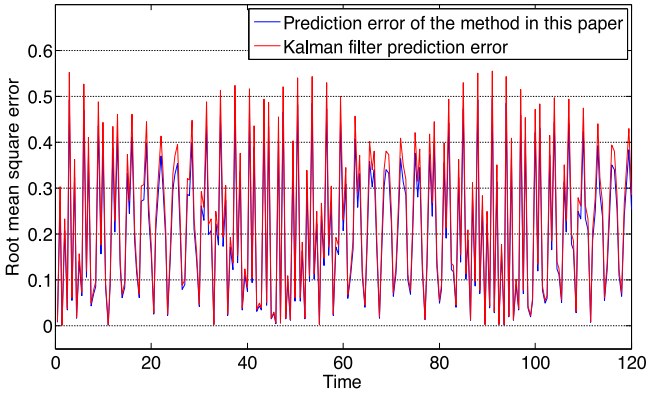


Fig. 10. Kalman filter and the root mean square error of the predicted and actual values.

and fully guarantee the response rate of the CPS system to emergencies, we analyze the complexity of the CPS system from the perspective of information theory. We set the threshold based on the average dynamic complexity of the CPS system to determine the level of information entropy of the sensor data in a certain period of time. The CPS system selects high-entropy data to send first, and analyzes its effectiveness through experiments. The analysis and guarantee of communication performance need not only the tasks of the physical layer and the data link layer, but also the cooperation of all relevant links and protocol stacks. Therefore, when a more general CPS network model is subsequently established and a resource allocation algorithm is proposed, it can start with a multi-layer cooperation mechanism and achieve the design goal of the multi-layer cooperation guarantee system.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Juxia Xiong: Data curation. **Jinzhao Wu:** Validation.

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Juxia Xiong was born in xiangyang Hubei province of china in 1982. she graduated from guangxi university for nationalities with an bachelor's degree in 2003. after graduating from guangxi university for nationalities with an master's degree in 2009 She worked as a teacher at guangxi university for nationalities. she has published 18 papers and holds 1 patent and 7 software copyrights. she became a doctoral student of chengdu institute of computer applications Chinese academy of sciences in 2015. her research interests include formal methods Automatic reasoning Data mining And intelligent computing. Xiongjuxia1107@163.com



Jinzhao Wu was born in Qiqihar, Heilongjiang Province of China in 1965. He graduated from the Institute of Systems of the Chinese Academy of Sciences with a PhD in 1994. He is a researcher and doctoral supervisor of the Chengdu Computer Application Institute of the Chinese Academy of Sciences. Now he works at Guangxi University for Nationalities. He has published more than 120 papers, holds 10 patents and 25 software copyrights. His research interests include formal methods, automatic reasoning, data mining, and intelligent computing. wjzgxun@163.com